

Haotian Cui, Ph.D.

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EDUCATION

07/2024	Ph.D., Computer Science Dissertation: Foundation Model for Single-Cell Omics	University of Toronto
06/2019	M.Sc., Biomedical Engineering	Tsinghua University
06/2015	B.Sc., Biomedical Engineering	Tsinghua University

EXPERIENCE

09/2025-	Senior AI Scientist	Xaira Therapeutics
02-08/2025	Postdoc Researcher Research on computational models for drug discovery. Coadvised by Prof. Bowen Li and Prof. Bo Wang	University of Toronto Vector Institute
07-12/2024	Worked on an AI-biotech Startup Creation Cofounder, led AI-driven drug discovery efforts and fund raising Full-stack AI for mRNA therapeutics, based on our works of scGPT, AGILE, etc.	
04-10/2024	Part-time research consultant Help build up the single-cell foundation model pipeline for vevo.ai Consulting for scGPT perturbation prediction on the internal assays for Bit.bio	
05-12/2021	Research Intern Advisor: Dr. Nan Duan, Dr. Jianfeng Gao Research on LLM pretraining for coding assistants, optimizing transformer-based models for enhanced code generation and understanding.	Microsoft Research
2019-2024	Graduate Student Researcher Advisor: Prof. Bo Wang Research on large-scale self-supervised models for single-cell omics and drug discovery	University of Toronto Vector Institute

RESEARCH INTERESTS

Machine Learning, Genomics, Drug Discovery

- Developing large-scale self-supervised models that enable discovery of biological insights and drug targets from single-cell omics..
- Foundation model and automatic laboratory for therapeutic discoveries, including mRNA drug optimization and delivery vehicle design.
- Multimodal learning for molecular cell biology, for discovery and understanding of generic regulations and multi-faceted developmental or pathological processes.

PUBLICATIONS (1000+ citations / [Google Scholar](#) / [GitHub](#))

FIRST/CO-FIRST AUTHORSHIP (*equal contribution, first three highlighted, sorted by year afterwards)

1. **Cui, H.***, Wang, C.*, Maan, H., Pang, K., Luo, F., Duan, N. and Wang, B., 2024. scGPT: toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods*, pp.1-11. [\[link\]](#)
 - Pioneering single-cell foundation model with generative pretraining, for building virtual cells.
 - Zero-shot capability of cell type calling and genetic target identification; fine-tuning for perturbation response prediction.
 - Highlighted by Nvidia GTC 2024 [\[link\]](#), NY Times [\[link\]](#), Nature Methods [\[link\]](#), [\[link\]](#),
2. **Cui, H.**, Tejada-Lapuerta, A., Brbić, M., Saez-Rodriguez, J., Cristea, S., Goodarzi, H., Lotfollahi, M., Theis, F.J., and Wang, B. 2025. Towards Multimodal Foundation Models in Molecular Cell Biology. *Nature*, 640(8059), pp.623-633. [\[link\]](#)
 - Perspective on techniques and data for a multimodal foundation model throughout the central dogma for building virtual cell
3. **Cui, H.***, Xu, Y.*, Pang, K., Li, G., Gong, F., Wang, B., Li, B., 2025. LUMI-lab: a Foundation Model-Driven Autonomous Platform Enabling Discovery of New Ionizable Lipid Designs for mRNA Delivery. *bioRxiv*, pp.2025-02. (Under review, *Cell*) [\[link\]](#), [\[video\]](#)

- Fully self-driving lab powered by a new molecular foundation model, conducting experiments in active-learning loops.
 - Discovered new family of ionizable lipids with brominated lipid tails for mRNA delivery
 - Achieved 20.3% gene editing efficiency in lung epithelial cells, the highest reported for inhaled LNP-mediated delivery in mice
- Wang, C.X.*, **Cui, H.***, Zhang, A.H., Xie, R., Goodarzi, H. and Wang, B., 2025. scGPT-spatial: Continual Pretraining of Single-Cell Foundation Model for Spatial Transcriptomics. *bioRxiv*, pp.2025-02.
 - Ying, K.*, Song, J.*, **Cui, H.***, Zhang, Y., Li, S., Chen, X., Liu, H., Eames, A., McCartney, D.L., Marioni, R.E. and Poganik, J.R., 2024. MethylGPT: a foundation model for the DNA methylome. *bioRxiv*, pp.2024-10.
 - Xu, Y.*, Ma, S.*, **Cui, H.***, Chen, J., Xu, S., Gong, F., Golubovic, A., Zhou, M., Wang, K.C., Varley, A. and Lu, R.X.Z., 2024. AGILE platform: a deep learning powered approach to accelerate LNP development for mRNA delivery. *Nature Communications*, 15(1), p.6305.
 - Cui, H.***, Maan, H.*, Vladiou, M.C., Zhang, J., Taylor, M.D. and Wang, B., 2024. DeepVelo: deep learning extends RNA velocity to multi-lineage systems with cell-specific kinetics. *Genome Biology*, 25(1), p.27.
 - Cui, H.**, Wang, C., Huang, J., Inala, J.P., Mytkowicz, T., Wang, B., Gao, J. and Duan, N., 2022, December. CodeExp: Explanatory Code Document Generation. In Findings of the Association for Computational Linguistics: *EMNLP 2022* (pp. 2342-2354).
 - Cui, H.***, Wang, C.*, Maan, H., Duan, N. and Wang, B., 2022. scFormer: a universal representation learning approach for single-cell data using transformers. *bioRxiv*, pp.2022-11.
 - Cui, H.***, Maan, H.*, Taylor, M.D. and Wang, B., 2022. A Deep Learning Framework for Estimating Cell-specific Kinetic Rates of RNA Velocity. *The 2022 ICML Workshop on Computational Biology*, 25(1), p.27.
 - Sugiyama, M.G.*, **Cui, H.***, Redka, D.Y.S., Karimzadeh, M., Rujas, E., Maan, H., Hayat, S., Cheung, K., Misra, R., McPhee, J.B. and Viirre, R.D., 2021. Multiscale interactome analysis coupled with off-target drug predictions reveals drug repurposing candidates for human coronavirus disease. *Scientific reports*, 11(1), p.23315.
 - Dou, Z.*, **Cui, H.***, Zhang, L. and Wang, B., 2020. Learning global and local consistent representations for unsupervised image retrieval via deep graph diffusion networks. *arXiv preprint arXiv:2001.01284*.

CO-AUTHORSHIP

- Szalata, A., Hrovatin, K., Becker, S., Tejada-Lapuerta, A., **Cui, H.**, Wang, B. and Theis, F.J., 2024. Transformers in single-cell omics: a review and new perspectives. *Nature Methods*, 21(8), pp.1430-1443.
- Tan, R., **Cui, H.**, Wang, B., Perez, K., Costa, A.D., Jordan, A., Karacic, T., Elganainy, D., Gui, D.Y., Kim, S. and Yuan, C., 2024. Abstract A029: Evaluating and interpreting scGPT: A foundation model for single-cell biology in real-world cancer clinical trial data. *Cancer Research*, 84(17_Supplement_2), pp.A029-A029.
- Hu, H., Ma, Y., Gao, X., Song, D., Li, M., Huang, H., Qian, X., Wu, R., Shi, K., Ding, H., Lin, M., Chen, X., Zhao, W., Qi, B., Zhou, S., Chen, R., Gu, Y., Chen, Y., Lei, Y., Wang, C., Wang, C., Tong, Y., **Cui, H.**, Abdal, A., Zhu, Y., Tian, X., Chen, Z., Lu, C., Yang, X., Mu, J., Lou, Z., Eghtedari, M., Zhou, Q., Oberai, A. and Xu, S., 2023. Stretchable ultrasonic arrays for the three-dimensional mapping of the modulus of deep tissue. *Nature Biomedical Engineering*, 7(10), pp.1321-1334.
- Zaidi, M., **Cui, H.**, Wang, B., McKee, T.D. and Wouters, B.G., 2021. Abstract PO-018: Computational staining of tumor hypoxia from H&E images using convolutional neural networks. *Clinical Cancer Research*, 27(5_Supplement), pp.PO-018.
- Huang, J., Wang, C., Zhang, J., Yan, C., **Cui, H.**, Inala, J.P., Clement, C., Duan, N. and Gao, J., 2022. Execution-based Evaluation for Data Science Code Generation Models. *DaSH 2022*, p.28.
- Yang, J., **Cui, H.**, Li, S., Gao, S., Guo, J. and Lu, Z., 2020. Outline Extraction with Question-Specific Memory Cells. *ACM Transactions on Asian and Low-Resource Language Information Processing (TALLIP)*, 19(4), pp.1-17.
- Liu, X., Mou, L., **Cui, H.**, Lu, Z. and Song, S., 2020. Finding decision jumps in text classification. *Neurocomputing*, 371, pp.177-187.
- Liu, X., Mou, L., **Cui, H.**, Lu, Z. and Song, S., 2018. Jumper: Learning when to make classification decisions in reading. *arXiv preprint arXiv:1807.02314*.

21. Li, G., Deng, L., Tian, L., **Cui, H.**, Han, W., Pei, J. and Shi, L., 2018. Training deep neural networks with discrete state transition. *Neurocomputing*, 272, pp.154-162.
22. Lu, Z., Liu, X., **Cui, H.**, Yan, Y. and Zheng, D., 2018, July. Object-oriented Neural Programming (OONP) for Document Understanding. In Proceedings of the 56th *Annual Meeting of the Association for Computational Linguistics* (Volume 1: Long Papers) (pp. 2717-2726).

PATENTS

- Self-Driving System and Method for AI-Based Design and Discovery of Lipid Nanoparticles as Therapy Delivery Vehicles (Provisional Application, USPTO 63/729,809)
- Systems and Methods for AI-Based Ionizable Lipid Development for Lipid Nanoparticle (LNP)-Based Cargo Molecule Delivery (PCT/CA2024/050644)

PROFESSIONAL SERVICE

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| 2019-now | Peer reviewer of journals and conferences, including Nature Methods , IEEE Transactions on Computational Biology and Bioinformatics , IEEE Transactions on Signal and Information Processing over Networks , Human Genetics , Bioinformatics , ACL , EMNLP |
| 2022 | Program committee member for ACL2022 DialDoc Workshop [link] |

PRESENTATIONS AND INVITED TALKS

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| 2025 | “Large models for single-cell omics and drug discovery: data, pretraining, and closed-loop environment”, RAAN Festival Satellite on AI in Genomics and Drug Discovery, Roche Inc. |
| 2024 | “scGPT: towards building a foundation model for single-cell multi-omics using generative AI”, Computational Biology Club, Vector Institute for Artificial Intelligence, Toronto |
| 2023 | “scGPT: towards building a foundation model for single-cell multi-omics using generative AI”, School of Medicine Single-Cell Club, New York University |
| 2023 | scGPT webinar, Superbio.ai |
| 2023 | “scGPT: towards building a foundation model for single-cell multi-omics using generative AI”, CEGS_ICA Journal Club, New York Genome Center |
| 2022 | “CodeExp: Explanatory Code Document Generation”, EMNLP 2022 |
| 2022 | “A Deep Learning Framework for Estimating Cell-specific Kinetic Rates of RNA Velocity”, The 2022 ICML Workshop on Computational Biology |

AWARDS AND HONORS

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| 2024 | Department of Computer Science Graduate Program Award , University of Toronto |
| 2022 | Best Poster Award , ICML 2022 Workshop on Computational Biology |
| 2014 | First Prize, Texas Instrument National Biomedical Electronics Competition |
| 2014 | Science and Technology Scholarship , Tsinghua University |
| 2013 | First Prize, NI National Virtual Instruments Competition |